2026 International Center for Polyamine Disorders Conference

July 13th and 14th, 2026
Grand Rapids, Michigan



Welcome to the second biennial International Center for Polyamine Disorders (ICPD) conference. We are delighted to have you here! This gathering is more than a series of talks – it's a special community coming together to share discoveries, ideas, and spark collaboration.

The 2026 International Center for Polyamine Conference is generously supported by the Corewell Health and Michigan State University Alliance Corporation.

We hope you will take every opportunity to learn, engage and get to know others with a shared commitment to furthering the polyamine community.

Here's to a few days of learning and inspiration!

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DAY 1 SCHEDULE

8:00 – 8:45 am	**Check-in & Breakfast**
8:45 – 9:15 am	**Welcome and Opening Remarks**
9:15 – 9:50 am	**From Rarity to Revolution: Redefining the Rare Disease Journey in the Age of AI** *Surender Rajasekaran, MD, MPH*
9:50 – 10:05 am	**Break**
10:05 – 10:40 am	**Tiny Molecules, Big Impact: Unlocking Early Diagnosis in Polyamine Disorders** *Sarah Elsea, PhD, FACMG*
10:40 – 11:15 am	**Demystifying Epilepsy – a Common Symptom of Neurogenetic Disorders** *Ryan Carrier, MD*
11:15 – 11:50 am	**Special Needs and Developmental Disability Planning** *Bobbi Hines*
11:50 – 12:50 pm	**Lunch**
12:50 – 1:25 pm	**Using Model Systems to Support Rare Disease Research** *Amy Wilstermann, PhD*
1:25 – 2:00 pm	**Small Molecules, Big Impact: How Metabolomics Helps Us Understand Health and Disease** *Maryam Goudarzi, PhD*
2:00 – 2:15 pm	**Break**
2:15 – 2:50 pm	**Development and Characterization of a Novel Zebrafish Model of Bachmann-Bupp syndrome** *Liz VanSickle, PhD*
2:50 – 3:25 pm	**Decoding Polyamine Disorders: A Multiomics Approach** *Liucija Vaicenavičiūtė*
3:25 – 4:00 pm	**All Attendee Group Photo** Location: outdoor space between CHP and Annex
4:00 – 6:00 pm	**Dinner**

2:00 – 3:25 pm

Optional:
Family Gathering

Outdoor space between Corewell Health Place and Annex

DAY 2 SCHEDULE

8:00 – 8:45 am	**Check-in & Breakfast**
8:45 – 9:20 am	**Policy, Progress, and Patients: Understanding the Decisions Shaping Rare Disease Care and Innovation** *Phil Yeiter*
9:20 – 9:55 am	**Fibroblasts Rely on Hypusination to Make Collagen Protein** *Ali McCubbrey, PhD*
9:55 – 10:30 am	**Development of Next-Generation Spermine Prodrugs for the Potential Treatment of Snyder-Robinson syndrome** *Luis Silva, PhD(c)*
10:30 – 10:45 am	**Break**
10:45 – 11:20 am	**The Rise of Virtual Cell Models for Rare Disease Research** *Zach deBruine, PhD*
11:20 – 11:55 am	**Beyond the Headlines: Navigating Emerging Treatments within a Healthcare System – *Education and awareness of clinical trial development for families seeking treatment options*** *Jodi Bohnhorst, MPA, ACRP-CP*
11:55 – 12:50 pm	**Lunch**
12:50 – 1:25 pm	**Making Health Better, Together** *Tina Freese Decker, MHA, MSIE, FACHE*
1:25 – 2:00 pm	**DNA Methylation Episignature Testing and its Application to Polyamine Disorders** *Matt Tedder, PhD*
2:00 – 2:35 pm	**Decoding Snyder-Robinson Syndrome: New Biological and Clinical Insights** *Tracy Murray Stewart, PhD*
2:35 – 2:50 pm	**Break**
2:50 – 3:25 pm	**Hepatic OAT Gene Replacement Normalizes Key Biochemical Abnormalities in a Mouse Model of Gyrate Atrophy** *Julia Johansson, PhD*
3:25 – 4:00 pm	**Polyamine Transporters: What They Are and Why They Matter** *Sarah Van Veen, PhD*
4:00 – 4:30 pm	**Closing Remarks**

2:35 – 4:00 pm

Optional:
Family Gathering

Outdoor space between Corewell Health Place and Annex

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Speaker Profiles



JODI BOHNHORST, MPA, ACRP-CP

Jodi G. Bohnhorst is the Director of Research Development at Corewell Health Research Institute in Grand Rapids, Michigan. She brings more than 17 years of experience in research administration, clinical trial development, and public health program leadership, with a focus on advancing innovative, patient and community-centered research initiatives. In her current role, Jodi leads statewide research development strategy, overseeing multidisciplinary teams that support grant proposal development and clinical trial start-up across diverse clinical areas. She partners closely with clinical leaders, investigators, and community stakeholders to enhance research access, streamline processes, and ensure regulatory and operational excellence. Jodi's career includes significant experience supporting federally funded initiatives, including work on large-scale genetics and public health collaborations such as the Region 4 Midwest Genetics Collaborative. Her work has emphasized strengthening connections between clinicians, researchers, and families to improve health outcomes and expand access to research participation. She is certified by the Association of Clinical Research Professionals (ACRP) and holds a Master of Public Administration from Grand Valley State University. Her leadership is grounded in a commitment to collaboration, community impact, and advancing research that reflects the needs of patients and families.



RYAN CARRIER, MD

Ryan Carrier, MD is a pediatric neurologist at Helen DeVos Children's Hospital. He completed his residency in pediatric neurology as well as fellowship in neurogenetics at the University of Rochester.

He has a special interest in genetic conditions with neurologic symptoms. He evaluates and manages a wide range of neurologic disorders including epilepsy, headaches, developmental/cognitive delays, and movement disorders.



ZACH DEBRUINE, PHD

Zach DeBruine is Assistant Professor of AI and Bioinformatics at Grand Valley State University. He completed his doctoral and postdoctoral training in high-performance bioinformatics at Van Andel Institute. His research lab has been supported by the Chan Zuckerberg Initiative and NIH and focuses on large-scale data engineering and foundation model training for genomics data that enable Virtual Cell tooling. Ultimately, these models aim to provide personalized insights into complex rare disease cases.



SARAH ELSEA, PHD, FACMG

Dr. Sarah H. Elsea is a Professor in Molecular & Human Genetics and the School of Health Professions, and Director of Clinical Genomics in the BCM-Human Genome Sequencing Center at Baylor College of Medicine. She is nationally and internationally recognized as an expert in neurodevelopmental and neurometabolic disorders. She co-developed clinical metabolomics at the BCM Medical Genetics Laboratories, now Baylor Genetics. As the Director of Clinical Genomics for the BCM-HGSC Clinical Laboratory, she oversees clinical reporting for the CLIA laboratory including large-scale research population genome sequencing projects, undiagnosed rare disease programs, ethical compliance, and integration of multi-omics data for diagnosis. Dr. Elsea's research program is focused on discovery, diagnosis, pathomechanisms, and treatment of rare disease across the lifespan, particularly neurodevelopmental, neurometabolic, and neurodegenerative disorders, from basic science to patient and family outcomes. Her clinical work at BCM focuses on development of clinical metabolomics, biomarker discovery, and approaches to treatment, as well as the utilization and integration of multi-omics technologies to develop and define model systems for investigation, treatment, and monitoring of both rare and complex disease.



TINA FREESE DECKER, MHA, MSIE, FACHE

Tina Freese Decker is President & CEO of Corewell Health, a leading integrated health system that provides health care and health insurance coverage to millions of people. She leads more than 60,000 team members who are passionate about making health better, together.

Tina is determined to tackle the health field's greatest challenges. She influences and shapes health policy, engages in advocacy, and builds strong relationships with decision makers across Michigan and beyond through engagement in national, state and community boards She is passionate about building culture and developing people to achieve their fullest potential. Tina inspires through innovation and collaboration to make health better for all.



MARYAM GOUDARZI, PHD

Maryam Goudarzi, PhD, is Associate Professor of Biochemistry and Molecular Biology and Director of the Mass Spectrometry and Metabolomics Core at Michigan State University. She is a metabolomics scientist whose research focuses on understanding how small molecules, known as metabolites, influence health and disease. Using advanced mass spectrometry technologies, her work investigates how metabolism changes in response to disease, inflammation, diet, and therapeutic interventions.

Dr. Goudarzi has contributed to studies spanning inflammatory bowel disease, microbiome-host interactions, cancer metabolism, and biomarker discovery. She also develops computational tools that help researchers interpret complex metabolomics data and translate molecular measurements into biological insights. Through her research and collaborations with scientists and clinicians, she works to advance precision medicine and improve our understanding of the molecular processes that drive human disease.



BOBBI HINES

Bobbi S. Hines is a Member in Miller Johnson's Probate and Estate Planning practice group, representing individuals and families in complex estate, elder law, special needs, Medicaid, and Veteran's aid planning and administration. With 15+ years of experience, she brings a practical, collaborative approach shaped by her personal journey as a teen mother who emancipated at 16, worked full-time through high school (graduating in the top 10), and pursued college as a nontraditional student while raising a child. Now married with three children, she is deeply passionate about special needs planning.

Ms. Hines assists clients with comprehensive estate planning, including wills, revocable/irrevocable trusts, special needs trusts, ILITs, succession planning, and probate/trust administration. She also provides elder law guidance on long-term care, powers of attorney, Medicaid applications, asset protection, and appeals. A member of the State Bar of Michigan, National Academy of Elder Law Attorneys, and Academy of VA Pension Planners, she has been recognized by The Best Lawyers in America for Trusts and Estates Law since 2023. She frequently presents community workshops on estate and long-term care planning.



JULIA JOHANSSON, PHD

Julia Johansson is a postdoctoral research fellow in the department of Ophthalmology at Johns Hopkins. She completed her undergraduate degree in behavioral neuroscience at the University of Delaware in 2018. After graduating, she worked as a research assistant in the Gene Therapy Program under the guidance of Dr. James Wilson at the University of Pennsylvania developing AAV gene therapies for CNS-related diseases. In 2026, she earned her PhD in Cellular and Molecular Medicine from Johns Hopkins in the lab of Dr. Jefferson Doyle, where she will be continuing her postdoctoral work developing therapies for the rare genetic disorder, Gyrate Atrophy.



ALEXANDRA MCCUBBREY, PHD

Dr. Alexandra McCubbrey is an Associate Professor of Medicine at National Jewish Health in Denver, Colorado. She received her PhD in Immunology from the University of Michigan in 2014. Her lab studies how dying cells instruct nearby living cells to initiate repair processes, which her lab has found to involve polyamines.



TRACY MURRAY STEWART, PHD

Dr. Tracy Murray Stewart is an Assistant Professor of Oncology at the Sidney Kimmel Comprehensive Cancer Center at Johns Hopkins. Her laboratory focuses on understanding the polyamine metabolic pathway in both healthy and disease states, with the goal of exploiting its self-homeostatic control mechanisms as a means for therapeutic intervention. In addition to cancer, Dr. Stewart is recognized for expertise in the rare X-linked neurodevelopmental condition Snyder-Robinson syndrome (SRS). She, along with Dr. Robert Casero, Jr. and their team at Johns Hopkins University, were recognized by the Snyder-Robinson Foundation as the 2020 Outstanding Researchers for key research that has resulted in a better understanding of polyamine ratios in SRS patient cells. Importantly, they discovered that the polyamine imbalance characteristic of SRS can be corrected by agents that utilize the polyamine transport system, expanding potential treatment options to include difluoromethylornithine (DFMO), an existing, FDA-approved drug, alone or in combination with a spermine mimetic. Her laboratory continues to investigate this potential in addition to further understanding the molecular mechanisms underlying the disease phenotype.



SURENDER (RAJA) RAJASEKARAN, MD, MPH

Surender “Raja” Rajasekaran, MD, MPH, is a pediatric critical care physician at Helen DeVos Children’s Hospital and Medical Director of Research at Corewell Health, West. He also serves as Chair of the Corewell Health–Michigan State University (CH MSU) Alliance Board of Directors. His research focuses on leveraging genomics, systems biology and translational science to develop life-saving therapies and improve outcomes for critically ill patients, particularly in those with complex and rare diseases. His work seeks to define how genomic variation, immune dysregulation, infections and metabolic dysfunction contribute to life-threatening illness. By integrating transcriptomics, metabolomics, and computational modeling, his research aims to identify biologically actionable pathways that can be translated into individualized therapies and improved patient outcomes.

Dr. Rajasekaran holds the Endowed DeVos Chair of Pediatric Research and leads initiatives in research development, mentorship, and education. He has established key collaborations with Michigan State University and the Van Andel Research Institute and contributes to NIH-funded research.



LUIS SILVA, PHD(c)

Luis Silva is a PhD candidate in Biomedical Sciences at the University of Central Florida, where his research in Dr. Otto Phanstiel IV's laboratory focuses on polyamine metabolism and therapeutic development for Snyder-Robinson Syndrome. He develops redox-responsive spermine prodrugs designed to restore intracellular spermine levels and correct polyamine imbalance in SRS. His work has demonstrated therapeutic efficacy in patient-derived fibroblasts and Drosophila models, identifying next-generation lead compounds for preclinical development. Luis has also contributed to research in cancer metabolism and metabolic therapeutics. He received the Best Poster Award at the 2025 Gordon Research Conference on Polyamines.



MATT TEDDER, PHD

Matthew graduated from Clemson University in 2018 with a PhD in Healthcare Genetics. Matthew is a staff scientist at the Greenwood Genetic Center (GGC) where his primary responsibilities include data review, variant classification, and managing EpiSign testing, both from a clinical and research perspective.



ELIZABETH VANSICKLE, PHD

Liz VanSickle is a scientist at the International Center for Polyamine Disorders and in the Division of Genetics and Genomics at Helen DeVos Children's Hospital. She holds a MS in Clinical Laboratory Sciences from Michigan State University and a PhD in Healthcare Genetics and Genomics from Clemson University. Prior to joining the ICPD in 2020, she worked as a scientist in pediatric oncology for 7 years at Helen DeVos Children's Hospital supporting phase I and II clinical trials for DFMO treatment of neuroblastoma and other pediatric solid tumors.



LIUCIJA VAICENAVIČIŪTĖ

Liucija Vaicenaviciute is a Bioinformatics Scientist at VUGENE, a biomedical data science company specializing in personalized multi-omics analysis and interpretation. She holds a Master's degree in Basic and Translational Neuroscience from the University of Cambridge. Bridging advanced data science with a deeply collaborative approach at VUGENE, Liucija leverages her interdisciplinary background to transform complex raw data into meaningful, actionable insights, ensuring researchers have the dedicated biological understanding needed to drive breakthroughs.



SARAH VAN VEEN, PHD

Sarah van Veen, PhD, is a molecular neurobiologist studying polyamine transport and its role in human health and disease. She obtained her MSc and PhD in Biomedical Sciences from KU Leuven (Belgium), where she currently works as a postdoctoral researcher in the Laboratory of Cellular Transport Systems. In August 2026, she will join the University of Cambridge (UK) as a Marie Skłodowska-Curie Postdoctoral Fellow. Her research focuses on how cells transport polyamines and how defects in these pathways contribute to neurodevelopmental and neurodegenerative disorders. Her work helped identify ATP13A2 and ATP13A4 as polyamine transporters, providing new insights into brain development and disease. At the ICPD meeting, she will discuss polyamine transporters and why they matter for understanding polyamine-related disorders.



AMY WILSTERMANN, PHD

Amy Wilstermann, PhD, is a Professor of Biology and Director of the Honors Program at Calvin University. She completed her graduate studies in biochemistry at Vanderbilt University and taught at a small undergraduate institution in Nashville, TN prior to beginning her work at Calvin. Amy uses yeast model systems and computational models to study mitochondrial rare diseases, particularly those caused by mutations in *BCS1L.* In her role as a founding member of the Rare Disease Network, she has the opportunity to work with collaborators to bring members of the rare disease community together to learn from and support one another.



PHIL YEITER

Phil Yeiter is the senior director of government relations at Corewell Health, a Michigan-based non-profit health system with 20+ hospitals and an affiliated health plan, Priority Health, covering 1.3 million lives. In this role he leads the approach to legislative and governmental interaction at the local, state and national levels that help shape the Corewell Health government relations strategy. Phil works with policy makers, business partners, and diverse organizations seeking to create and influence legislation to ensure our patients receive the right care, at the right time, and at the right place. Phil's areas of expertise include lobbying, campaign finance, contract management, relationship building, policy and budget development, compliance, and process improvement.

Prior to joining Corewell Health, Phil was the legislative and policy liaison at the State of Michigan Department of Military and Veterans Affairs, where he oversaw the development and execution of legislative policies aimed at addressing issues facing Michigan's veteran population and their families; specializing in health care and long-term care facility administration. Prior to that, Phil held several roles in the Michigan state legislature, most notably as legislative advisor to the Chair of the House Appropriations Committee. Phil served seven years in the United States Army as an intelligence analyst, having deployed to Baghdad, Iraq in support of Operation Iraqi Freedom during 2004 to 2006.